Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Foster, Russell Grant Ph.D.		
Project Title:	The Role and Characterization of Novel Photoreceptor Mechanisms Regulating Circadian Rhythms, Sleep, Body Temperature and Heart Rate		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	UNITED KINGDOM		
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	01/01/2004	**End Date:**	12/31/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-HPF00301		
Performance Goal No.:			
Performance Goal Text:			
	(1) The original aims of the project: The long-term aims of this project are to provide effective countermeasures for the detrimental effects of the irregular and altered patterns of light exposure in space and on the surface of other planets. These countermeasures will depend upon an understanding of the basic mechanisms of how the eye perceives light. All photoreception within the eye of mammals was considered the province of the rods and cones. However, our recent studies have shown that varied aspects of physiology and behavior are regulated by gross changes in environmental light and that these irradiance changes are detected by non-rod, non-cone ocular photoreceptors. In 2003 we were able to demonstrate in the mouse that a small number (~ 1%) of retinal ganglion cells are directly photosensitive to light. It is these cells, termed photosensitive retinal ganglion cells (pRGCs) that seem to play a critical role in regulating varied aspects of our physiology and behavior. The central aims of the research that we are undertaking are to characterize the molecular		

Task Description:	mechanism of this unexplored photoreceptor system of the eye, and determine the extent to which the pRGCs contribute to varied aspects of physiology and behavior. In this regard our proposal spans 1-4 of the Countermeasures Readiness Levels (CRL) of NASA's Biomedical Research and Countermeasures Program. By employing a unique rodless and coneless mouse model (rd/rd cl), and by taking advantage of the new opportunities created by a range of post-genomic technologies, we will address two broad questions: Question 1: What is the role of pRGCs in the regulation of general physiology and behavior? Question 2: What are the molecular mechanisms of non-rod, non-cone ocular photoreceptors? The results from these integrative studies will provide the mechanistic substrate for both targeted drug development aimed at the manipulation of human circadian rhythms, sleep, mood and performance, and the design of new lighting sources that are either highly effective in regulating these novel photoreceptors or leave them largely unstimulated. (2) The key findings to-date: (i) Light detection by the eye is mediated by rods, cones and photosensitive retinal ganglion cells (pRGCs). The photopigment of the pRGCs has been the subject of considerable debate. From our previous data, the melanopsin gene and its protein product emerged as the strongest candidate for this photopigment. As part of our collaborative interactions we have been able to show, using gene expression approaches, that melanopsin forms the photopigment of the pRGCs. These data were published in Nature. (ii) Light detection by the eye is mediated by rods, cones and photosensitive retinal ganglion cells (pRGCs), but it has remained unclear which of these photoreceptor classes modulate sleep and alertness. Our data provide the first direct physiological and behavioural evidence that pRGCs modulate sleep. Furthermore we show that mice which have rods and cones but lack melanopsin no longer show sleep induction in response to light exposure. Thus the pRGCs provide either the dominant or exclusive photic input into the sleep switch of the brain. This work is being prepared for a submission to Nature. (iii) Cardiovascular performance, blood pressure and core body temperature can be modified by changes in ambient light. However, these responses have only been poorly characterised. Our on-going studies suggest that the acute regulation of heart rate by environmental light in animal models arises from inputs from a time-dependent interaction between the classical rod and cone photoreceptor pathways and the pRGCs. (iv) To investigate the molecular mechanisms where by light is transduced into a physiological response in pRGCs we have undertaken a microarray project looking at light induced gene expression. Our results show that approximately 20% of the transcriptome of the eye is regulated by pRGCs and we have identified a large number of candidate genes/proteins involved in the phototransduction cascade of these neurons. (3) The impact of these findings on the aims of the project: When our project was originally reviewed support was enthusiastic but concerns were raised that we had been overly ambitious. We feel that in the last 12 months we have fulfilled an impressive set of goals and that as we enter the final funding period of this application, we are in a very strong position to fulfil all the original aims of the project, and are ready to build upon our findings in a future application to NSBRI. In brief, we have been able to show that the photic stimulation of pRGCs must be considered the primary target for the development of effective countermeasures for the detrimental effects of the irregular and altered patterns of light exposure in space and on the surface of other planets. (4) The proposed research plan for the coming year: We have four major goals for the coming 12 months. a) The publication of the sleep results in Nature or allied high-impact journal; b) The completion of the data-set relating to the photic modulation of cardiovascular performance, blood pressure and core body temperature and the submission of this work for publication; c) Placing the genes identified in our microarray experiments into a functional context by using in situ localisation followed by gene ablation approaches; d) The preparation of a new NSBRI application to seek funds to complete the dissection of the melanopsin phototransduction pathway, and to develop and test countermeasures for the detrimental effects of the irregular and altered patterns of light exposure using rodent models and human subjects.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	New approaches to address the clinical problems that either result from, or are affected by, inappropriate lighting conditions (e.g. depression, insomnia, reduced alertness and circadian maladjustment associated with shift work, blindness or jet-lag) will be critically dependent upon a detailed understanding of the pRGCs. Thus a major earth-based benefit of the work outlined in this research project will be to provide basic information for the design of new lighting sources that can be used to modulate mood and performance, including depression. Furthermore we will provide the mechanistic substrate for targeted drug development aimed at the manipulation of human circadian rhythms, sleep, mood and performance. Another benefit will be a greater understanding of the G-coupled receptor superfamily. This large family of membrane receptors is a major target for drug development, but many details of how these receptors work remain to be elucidated. A final benefit from this work will be in the field of retinal disease. It has been estimated that over half of blindness can be attributed to retinal genetic defects and the involvement of inner retinal dysfunction in promoting dystrophy is an area of growing interest. Consequently, the functions of novel genes within the inner retina that either cause or contribute to genetic pathologies of the eye are of considerable significance.
Task Progress:	When our project was originally reviewed support was enthusiastic but concerns were raised that we had been overly ambitious. We feel that in the last 12 months we have fulfilled an impressive set of goals and that as we enter the final funding period of this application, we are in a very strong position to fulfil all the original aims of the project, and are ready to build upon our findings in a future application to NSBRI. Collectively, our findings provide the first physiological evidence that melanopsin-based pRGCs regulate sleep; that the acute regulation of heart rate by environmental light arises from a time-dependent interaction between the classical rod and cone photoreceptor pathways and melanopsin-based pRGCs; and we have identified unique elements of the melanopsin phototransduction pathway. In short, we have been able to show that the photic stimulation of pRGCs must be considered the primary target for the development of effective countermeasures for the detrimental effects of the irregular and altered patterns of light

	exposure in space and on the surface of other planets.
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